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GEOGRAPHY • Ocean water: relief, temperature, salinity, waves, tides and currents

UNDERSTAND IT FIRST

What is it? Oceans cover about **71%** of the Earth. Their water is **never still**: **waves** ride on the surface, **tides** raise and lower the sea twice a day, and **currents** move water like rivers in the sea. These movements shape **climate (warm and cold coasts, deserts, fog), fishing grounds, ports and shipping**, and even India's **monsoon** (El Niño). Learn it as **oceans and floor relief → temperature and salinity → waves and tsunamis → tides → currents and their effects → coral reefs**.

1. Oceans, floor relief, temperature and salinity

- **Oceans:** the **Pacific** is the largest and deepest (the **Mariana Trench**, about **11 km** deep); then the **Atlantic**, the **Indian Ocean** (the only ocean **named after a country**) and the **Arctic** (smallest). Some also count the **Southern Ocean** around Antarctica.
- **Floor relief:** the **continental shelf** (shallow, up to about **200 m**; the best **fishing grounds** and **oil fields** such as **Mumbai High**; wider off India's **west coast**), the steep **continental slope**, the **continental rise**, the flat **abyssal plain**, **mid-ocean ridges** (underwater mountain chains, e.g. the **Mid-Atlantic Ridge**) and deep **trenches** where plates sink.
- **Temperature** falls with depth; the zone of sharp fall is the **thermocline**. **Salinity** (salt content) averages about **35 parts per thousand (‰)**; it is high where **evaporation is high and fresh water is low** (**Red Sea**, about **40‰**; inland lakes like the **Dead Sea** and **Lake Van** are far saltier) and low where **rivers and rain add fresh water** (**Baltic Sea**, about 7‰). The **Bay of Bengal** is **less salty** than the **Arabian Sea** because the **Ganga, Brahmaputra, Godavari and Krishna** pour fresh water into it.

Relief of the ocean floor

2. Waves and tides

- **Waves** are mostly made by **wind** blowing over the sea; the water moves **up and down in circles**, while the **energy** moves forward. The top is the **crest**, the low point the **trough**. A **tsunami** ('harbour wave' in Japanese) is a series of huge waves caused by an **undersea earthquake, landslide or volcano**; in deep water it is fast and low, but it **rises to great heights near the coast** (Indian Ocean tsunami, **2004**).
- **Tides** are the regular **rise (high tide) and fall (low tide)** of the sea, caused mainly by the **Moon's gravity** (the Sun's pull is about half as strong) along with the Earth's rotation. Most coasts get **two high and two low tides** in about **24 hours 52 minutes**. **Spring tides** (biggest range) come at **new moon and full moon**, when the Sun, Earth and Moon are **in a line**; **neap tides** (smallest range) come at the **first and third quarter**, when the Sun and Moon are at **right angles**.
- The **Bay of Fundy** (Canada) has one of the world's **highest tidal ranges** (about **16 m**). In India, the **Gulf of Khambhat** and **Gulf of Kutch** have large ranges, good for **tidal energy**. **Uses of tides:** they help ships enter

tidal ports (**Kolkata** on the Hooghly, **Kandla**), clear silt from river mouths, and aid fishing.

Spring tides and neap tides

3. Ocean currents

- **Ocean currents** are large, steady streams of water. **Causes: planetary winds**, differences in **temperature and salinity (density)**, the **Coriolis force** and the **shape of coastlines**. They form big loops (**gyres**) that turn **clockwise in the north** and **anticlockwise in the south**. **Warm currents** flow from the **equator towards the poles** (usually along **eastern coasts** of continents in low latitudes); **cold currents** flow from the **poles towards the equator** (usually along **western coasts**).

Ocean	Warm currents	Cold currents
Atlantic	**Gulf Stream** and **North Atlantic Drift** (keep **NW Europe's ports ice-free**, e.g. Norway); **Brazil**	**Labrador** (off NE Canada), **Canary**, **Benguela** (off SW Africa; **Namib desert**), **Falkland**
Pacific	**Kuroshio ('Japan current')**, **East Australian**	**Oyashio** (off Japan), **California**, **Humboldt or Peru** (off Peru and Chile; **Atacama desert**, rich **anchovy** fishing)
Indian	**Agulhas** (off SE Africa); the **monsoon currents reverse direction** with the seasons in the north Indian Ocean	**West Australian**

- **Effects:** warm currents **warm the coasts** and bring rain; cold currents make coasts **cool and dry**, which is why **coastal deserts** (Atacama, Namib) lie beside them. Where **warm and cold currents meet** (the **Grand Banks** near Newfoundland, where the Gulf Stream meets the Labrador; off **Japan**, where the Kuroshio meets the Oyashio), mixing brings up nutrients for **plankton**, making the **world's richest fishing grounds**, though also **thick fog**.
- **El Niño** is an unusual **warming of the eastern Pacific** off Peru (it weakens the cold Humboldt current); it often **weakens India's monsoon**. **La Niña**, the opposite cooling, often brings **good rains** to India. A **positive Indian Ocean Dipole (IOD)** (western Indian Ocean warmer than the eastern) **helps** India's monsoon.
- **Coral reefs** grow in **warm (20–28°C), clear, shallow** tropical seas: **fringing** reefs (close to the coast), **barrier** reefs (separated by a lagoon; the **Great Barrier Reef**, Australia) and **atolls** (ring-shaped, e.g. **Lakshadweep**). India's reefs: **Gulf of Mannar, Gulf of Kutch, Andaman and Nicobar, Lakshadweep**. Warmer seas cause **coral bleaching**.

EXAM TRAP

Oceans **71%**; **Pacific** largest, deepest (**Mariana ~11 km**); Atlantic; **Indian (named after a country)**; Arctic smallest; floor: **shelf (≤200 m; fishing, Mumbai High), slope, rise, abyssal plain, ridge, trench**; **thermocline**; salinity **~35‰**; high **Red Sea (~40‰)**, Dead Sea, Lake Van; low **Baltic**; **Bay of Bengal less saline than Arabian Sea**; waves by **wind**; crest, trough; **tsunami** (undersea quake; 2004); tides mainly by **Moon**; **2 high + 2 low in ~24 h 52 min**; **spring (new/full moon; in line)**, **neap (quarters; right angle)**; **Bay of Fundy ~16 m**; **Khambhat, Kutch** (tidal power); tidal ports **Kolkata, Kandla**; currents: winds, density,

Coriolis, coastlines; gyres **clockwise N**; warm: **Gulf Stream, N Atlantic Drift (Norway ice-free), Brazil, Kuroshio, E Australian, Agulhas**; cold: **Labrador, Canary, Benguela (Namib), Falkland, Oyashio, California, Humboldt (Atacama, anchovy), W Australian**; **Indian Ocean currents reverse with monsoon**; **Grand Banks, Japan** (mixing → fishing, fog); **El Niño** weakens monsoon; **La Niña**; **positive IOD helps**; corals: fringing, barrier (Great Barrier Reef), atoll (Lakshadweep); India reefs **Mannar, Kutch, A&N, Lakshadweep**; bleaching.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Spring tides occur at new moon and full moon.

2. Neap tides occur when the Sun and the Moon are at right angles to the Earth.

3. The Bay of Fundy has one of the highest tidal ranges in the world.

4. Tides are caused mainly by the gravitational pull of the Sun.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2, 3 and 4

D. 1, 2 and 3 only

Answer: D. The Moon's pull is the main cause; being much closer, its tide-raising force is about twice the Sun's.

PRACTICE

Match List-I (ocean current) with List-II (description):

List-I (ocean current)	List-II (description)
A. Gulf Stream	1. Warm current off Japan
B. Labrador	2. Cold current off Peru and Chile
C. Humboldt	3. Cold current off north-east Canada
D. Kuroshio	4. Warm current of the North Atlantic

(1) A → 3 | B → 4 | C → 2 | D → 1

(2) A → 4 | B → 3 | C → 2 | D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (2) Gulf Stream: warm Atlantic; Labrador: cold Canada; Humboldt: cold Peru; Kuroshio: warm Japan.

PRACTICE

Which one of the following statements is NOT correct?

A. In the tropics, warm currents usually flow along the western coasts of continents.

B. The Pacific is the largest ocean.

C. The average salinity of ocean water is about 35 parts per thousand.

D. The Bay of Bengal is less saline than the Arabian Sea.

Answer: A. In low latitudes, warm currents flow along eastern coasts and cold currents along western coasts.

PRACTICE

Assertion (A): The ports of Norway remain ice-free in winter.
Reason (R): The cold Labrador current flows along the coast of Norway.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: the warm North Atlantic Drift (from the Gulf Stream) keeps Norway's ports open; the Labrador current flows off Canada.

PRACTICE

The Grand Banks near Newfoundland and the seas off Japan are among the world's richest fishing grounds. What is the main reason?

A. They lie over deep ocean trenches

B. Warm and cold currents meet there, and the mixing feeds plankton

C. They are surrounded by coral atolls

D. Their water has very low salinity

Answer: B. The Gulf Stream–Labrador and Kuroshio–Oyashio meetings bring up nutrients, so plankton and fish thrive.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the oceans, floor relief, thermocline and salinity facts. 2. Write waves, tsunamis and tides (causes, spring, neap, uses). 3. Write the causes of currents, the warm and cold list by ocean, their effects, El Niño, IOD and coral reefs.

ECONOMY · Strategy of planning in India; NITI Aayog

UNDERSTAND IT FIRST

What is it? Economic planning means the government sets **targets and priorities** for a period (usually **five years**) and decides **where to invest**. After 1947, India chose **planned development** in a **mixed economy**, with the **public sector** leading heavy industry. Twelve **Five-Year Plans** ran from **1951 to 2017**. After the **1991 reforms**, planning became **indicative** (guiding, not commanding), and in **2015** the **Planning Commission** was replaced by **NITI Aayog**, a **think tank** that works with the States. Learn it as **early ideas → Planning Commission and NDC → the twelve Plans → strategy changes → NITI Aayog → comparison**.

Planning ideas before 1950

- **M. Visvesvaraya** wrote **'Planned Economy for India' (1934)**. The Congress set up the **National Planning Committee (1938)** under **Nehru** (when **Subhas Chandra Bose** was Congress president). The **Bombay Plan (1944)** was written by industrialists such as **J.R.D. Tata and G.D. Birla**; the **Gandhian Plan (1944, S.N. Agarwal)**, the **People's Plan (1945, M.N. Roy)** and the **Sarvodaya Plan (1950, Jayaprakash Narayan)** followed. The **Planning Commission** was set up on **15 March 1950** by a **Cabinet resolution**: it was **neither constitutional nor statutory**; the **Prime Minister** was its **chairman**. The **National Development Council (NDC, 6 August 1952)**, with the PM and all **Chief Ministers**, **approved** the Plans.

1. The Five-Year Plans

Plan	Focus and facts
1st (1951–56)	**Agriculture**, irrigation and dams (Bhakra–Nangal, Hirakud); **Harrod–Domar** model; growth beat its target; **Community Development Programme (1952)**.

2nd (1956–61)	**Heavy industry** led by the public sector; **P.C. Mahalanobis model**; steel plants at **Bhilai, Durgapur, Rourkela**; Industrial Policy Resolution 1956.
3rd (1961–66)	**Self-reliance** and agriculture; **failed** because of the **wars of 1962 (China) and 1965 (Pakistan)** and **droughts**. Then **Plan holidays (1966–69)** with three **annual plans**, when the **Green Revolution** began.
4th (1969–74)	**Growth with stability** and self-reliance; **bank nationalisation (1969)**; the **'Garibi Hatao'** slogan (1971).
5th (1974–79)	**Removal of poverty** and self-reliance; **ended a year early (1978)** by the Janata Government, which brought a **Rolling Plan (1978–80)**.
6th to 7th (1980–90)	**6th**: poverty and jobs (**IRDP**); **7th**: **'food, work and productivity'**. Then **annual plans (1990–92)** during the crisis.
8th to 10th (1992–2007)	**8th**: after the **1991 reforms**; **human development**; **indicative planning**. **9th**: **growth with social justice and equity**. **10th**: **8% growth** target and **monitorable targets** for poverty, literacy and health.
11th and 12th (2007–17)	**11th**: **'faster and more inclusive growth'**. **12th**: **'faster, more inclusive and sustainable growth'**, the **last** Five-Year Plan.

- **Strategy shift:** till 1991, planning was close to **imperative** (the State decided investment), aimed at **import substitution**, **heavy industry** and a big **public sector**, with **licences** controlling private firms. Results: a strong industrial base, dams, **IITs**, self-sufficiency in food after the Green Revolution; but **slow growth ('Hindu rate')**, inefficiency, persistent poverty and a **balance-of-payments crisis (1991)**. After 1991, plans became **indicative**, letting the **market and private sector** lead.

2. NITI Aayog

NITI Aayog at a glance

- The **National Institution for Transforming India (NITI) Aayog** was set up on **1 January 2015**, also by a **Cabinet resolution** (not constitutional or statutory). **Chairperson: the Prime Minister**; a **Vice-Chairperson** (the first was **Arvind Panagariya**), full-time members, up to four Union Ministers as **ex-officio members**, special invitees and a **CEO**. The **Governing Council** includes **all Chief Ministers** and Lieutenant Governors; **Regional Councils** deal with regional issues.
- **Role:** a **policy think tank** promoting **cooperative federalism** (working with States) and **competitive federalism** (ranking States on **indices**: **SDG India Index**, **Composite Water Management Index**, **School Education Quality Index**, **Health Index**, **Export Preparedness Index**, **India Innovation Index**, **Fiscal Health Index**, the **national MPI**). It does **not allocate funds** (that went to the **Finance Ministry**). Programmes: **Atal Innovation Mission (2016)** with **Atal Tinkering Labs**; the **Aspirational Districts Programme (2018, 112 districts**, including **Vizianagaram, Visakhapatnam and YSR Kadapa** in AP**)**; the **Aspirational Blocks Programme (2023, 500 blocks)**; the **Development Monitoring and Evaluation Office (DMEO)**; work on the **'Viksit Bharat @2047'** vision.

Point	Planning Commission	NITI Aayog
Set up	15 March 1950	1 January 2015
Approach	**Top-down**; one plan for all	**Bottom-up**; cooperative and competitive federalism
Funds	**Allocated** plan funds to ministries and States	**No** power to allocate funds; advisory
States' role	Through the NDC; limited	**Governing Council** of all CMs; greater
Plans	**Five-Year Plans** (1951–2017)	No Five-Year Plans; vision and strategy documents

EXAM TRAP

Visvesvaraya 1934; **NPC 1938 (Nehru)**; **Bombay Plan 1944**; **Gandhian 1944 (S.N. Agarwal)**; **People's 1945 (M.N. Roy)**; **Sarvodaya 1950 (JP)**; **Planning Commission 15 Mar 1950** (Cabinet resolution; PM chair); **NDC 6 Aug 1952** (approves); **1st: agriculture, Harrod–Domar**; **2nd: heavy industry, Mahalanobis, Bhilai–Durgapur–Rourkela**; **3rd: failed (1962, 1965 wars, drought)**; **plan holidays 1966–69** (Green Revolution); **4th: growth with stability, bank nationalisation 1969**; **5th: poverty removal, ended 1978**; **Rolling Plan 1978–80**; **6th: IRDP**; **7th: food, work, productivity**; **8th: indicative, human development**; **9th: social justice**; **10th: 8%, monitorable targets**; **11th: inclusive**; **12th: sustainable, last**; imperative → indicative; **NITI 1 Jan 2015**; PM chair; first VC **Panagariya**; **Governing Council (all CMs)**; cooperative + competitive federalism; indices; **no fund allocation**; **AIM 2016**; **Aspirational Districts 2018 (112; AP: Vizianagaram, Visakhapatnam, YSR Kadapa)**; **Aspirational Blocks 2023 (500)**; DMEO; Viksit Bharat @2047.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Planning Commission:
1. It was set up in March 1950 by a resolution of the Union Cabinet.
2. It was a constitutional body.
3. The Prime Minister was its chairman.
4. Five-Year Plans were finally approved by the National Development Council.
Which of the above are correct?

A. 1 and 3 only

B. 2, 3 and 4 only

C. 1, 3 and 4 only

D. 1, 2, 3 and 4

Answer: C. The Planning Commission was an extra-constitutional, non-statutory body.

PRACTICE

Match List-I (Plan) with List-II (focus):

List-I (Plan)	List-II (focus)
A. First Plan	1. Faster and more inclusive growth
B. Second Plan	2. Removal of poverty; ended a year early
C. Fifth Plan	3. Mahalanobis model; heavy industries
D. Eleventh Plan	4. Agriculture; Harrod–Domar model

(1) A → 4 B → 3 C → 2 D → 1

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (1) 1st: agriculture; 2nd: Mahalanobis; 5th: poverty, ended 1978; 11th: inclusive growth.

PRACTICE

Which one of the following statements about NITI Aayog is NOT correct?

A. It was set up on 1 January 2015.

B. Its Governing Council includes all Chief Ministers.

C. It runs the Aspirational Districts Programme launched in 2018.

D. It allocates plan funds to the States, as the Planning Commission did.

Answer: D. NITI Aayog is an advisory think tank; it has no power to allocate funds.

PRACTICE

Assertion (A): India had annual plans instead of a Five-Year Plan from 1966 to 1969.

Reason (R): The Green Revolution began in the late 1960s.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but the plan holiday was caused by the failure of the Third Plan (wars, droughts), not by the Green Revolution.

PRACTICE

A State Government wants to see how it ranks against other States on progress towards the Sustainable Development Goals. Which publication should it consult?

A. The Finance Commission's report

B. NITI Aayog's SDG India Index

C. The RBI Annual Report

D. The CAG's audit report

Answer: B. NITI Aayog's SDG India Index ranks States and UTs on the SDGs.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the pre-1950 plans, the Planning Commission and the NDC. 2. Write the 12 Plans with focus, plus plan holidays and the Rolling Plan, and the strategy shift. 3. Write NITI Aayog's structure, role, indices and programmes, and the comparison table.

SCIENCE · Environmental pollution: types, sources and impacts

UNDERSTAND IT FIRST

What is it? Pollution is the entry of **harmful substances or energy** (pollutants) into air, water, soil or surroundings that **harms living things or property**. India has some of the world's **most polluted cities**, rivers with high sewage loads, and areas with **fluoride or arsenic** in groundwater. Exams ask about **pollutants, their sources, health effects, measuring terms (AQI, BOD), diseases, laws and Indian cases**. Learn it as **air → water → soil → noise → thermal, radioactive, light and marine → laws and bodies → AP cases**.

1. Air pollution

Point	Details

Pollutants	**Primary** (released directly): **particulate matter (PM10, PM2.5)**, **sulphur dioxide (SO_2)**, **nitrogen oxides (NOx)**, **carbon monoxide (CO)**, volatile organic compounds, lead. **Secondary** (formed in the air): **ground-level ozone** and **PAN** (from NOx and hydrocarbons in **sunlight**), **acid rain**. **PM2.5** (smaller than **2.5 micrometres**) goes deep into the **lungs and blood**. **CO** binds to **haemoglobin** and cuts oxygen supply.
Sources	**Vehicles**, industries and **coal power plants**, **construction and road dust**, **burning of crop stubble** (Punjab and Haryana in **October–November**), waste burning, firecrackers, and indoor **chulhas** using wood and dung (cut by **Ujjwala** LPG).
Smog and acid rain	**Classical (sulphurous) smog**: smoke + fog in cold, damp air (**London, 1952**). **Photochemical smog**: NOx and hydrocarbons react in **sunlight** to form ozone (**Los Angeles**; Delhi). **Acid rain** (**pH below 5.6**) comes from **SO_2 and NOx**; it damages lakes, forests and buildings; the yellowing of the **Taj Mahal** ('**marble cancer**') led to the **Taj Trapezium Zone** (Supreme Court, **M.C. Mehta, 1996**).
Measuring and control	**National Ambient Air Quality Standards (2009)** cover **12 pollutants** (PM2.5 annual limit **40 µg/m³**, while the **WHO 2021 guideline is 5 µg/m³**). The **National Air Quality Index (2014)** uses **8 pollutants** and **6 categories**. The **National Clean Air Programme (NCAP, 2019)** covers about **130 non-attainment cities**, aiming to cut particulate levels by up to **40% by 2025–26** (from 2017 levels). In Delhi–NCR, the **Commission for Air Quality Management (2021)** enforces the **Graded Response Action Plan (GRAP)**. The **Bhopal gas tragedy (1984)** was caused by **methyl isocyanate (MIC)**.

National Air Quality Index categories

2. Water and soil pollution

- **Sources:** untreated **sewage**, **industrial effluents**, **fertiliser and pesticide** run-off, and **aquaculture** waste. **Measures: Dissolved Oxygen (DO)**: fish need about **4–5 mg/L** or more; **Biochemical Oxygen Demand (BOD)**: oxygen microbes need to break down organic waste; **higher BOD means more pollution** and less oxygen for fish; **COD** measures all oxidisable matter; **faecal coliform** bacteria show sewage. Excess **nitrates and phosphates** cause **eutrophication** (Day 25).
- **Diseases from chemicals: mercury → Minamata disease** (Japan); **cadmium → itai-itai** ('ouch-ouch') disease; **nitrates → blue baby syndrome**; excess **fluoride → fluorosis** (teeth and bones; found in **Prakasam** and parts of **Rayalaseema**, as well as Nalgonda in Telangana); **arsenic → skin lesions and cancer** (Ganga–Brahmaputra plains, West Bengal); **lead** → brain damage in children. **Soil pollution** comes from overuse of **chemicals**, **industrial dumping**, plastics and **salinity** from poor irrigation. The **Namami Gange Programme (2014)** cleans the Ganga.

3. Other types

Type	Facts
Noise	Measured in **decibels (dB)**; long exposure above about **85 dB** can damage hearing. The **Noise Pollution (Regulation and Control) Rules, 2000** set day/night limits: **industrial 75/70**, **commercial**

	65/55, **residential 55/45**, **silence zone 50/40 dB(A)**; a **silence zone** covers **100 m** around **hospitals, schools and courts**; loudspeakers are not allowed from **10 p.m. to 6 a.m.** (with limited exceptions).
Thermal	**Hot water** released by **power plants and industries** into rivers and the sea **lowers dissolved oxygen** and kills fish.
Radioactive	From **nuclear accidents** (**Chernobyl 1986**, **Fukushima 2011**), nuclear waste and testing; causes **cancer and genetic damage**. Natural background radiation is high on Kerala's **monazite** beaches.
Light and marine	**Light pollution** confuses birds, insects and **sea-turtle hatchlings**. **Marine pollution**: **oil spills**, **plastics** and **microplastics**, sewage, and **'dead zones'** with no oxygen.

- **Laws and bodies: Water (Prevention and Control of Pollution) Act, 1974** set up the **Central and State Pollution Control Boards**; **Air Act, 1981**; **Environment (Protection) Act, 1986**; **National Green Tribunal (2010)**. **AP cases:** the **LG Polymers styrene gas leak** at **Visakhapatnam (7 May 2020)**; pollution of **Kolleru** by fish tanks (cleared in **'Operation Kolleru', 2006**); **fluoride** in groundwater in **Prakasam** and Rayalaseema; industrial pollution near **Visakhapatnam** and **Kakinada**.

EXAM TRAP

Primary: **PM10, PM2.5, SO_2, NOx, CO**, VOCs, lead; secondary: **ground ozone, PAN, acid rain**; **PM2.5 < 2.5 µm**; CO + haemoglobin; sources: vehicles, coal plants, dust, **stubble (Oct–Nov)**, chulhas (**Ujjwala**); **London smog 1952 (sulphurous)**; **photochemical (LA)**; acid rain **pH < 5.6**; **Taj 'marble cancer'**, **TTZ (M.C. Mehta 1996)**; **NAAQS 2009 (12)**; PM2.5 **40 vs WHO 5**; **AQI 2014 (8 pollutants, 6 categories)**; **NCAP 2019 (~130 cities, up to 40% by 2025–26)**; **CAQM 2021, GRAP**; **Bhopal 1984 (MIC)**; **DO ≥ 4–5 mg/L**; **high BOD = more pollution**; COD; coliform; **Minamata (Hg), itai-itai (Cd), blue baby (nitrate), fluorosis (F: Prakasam, Rayalaseema), arsenic (WB)**, lead; **Namami Gange 2014**; noise **dB**; limits **75/70, 65/55, 55/45, 50/40**; silence zone **100 m**; **10 p.m.–6 a.m.**; thermal (DO falls); **Chernobyl 1986, Fukushima 2011**; light, marine, microplastics, dead zones; **Water Act 1974 (CPCB, SPCBs), Air Act 1981, EPA 1986, NGT 2010**; AP: **LG Polymers 7 May 2020**, **Operation Kolleru 2006**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. PM2.5 refers to particles smaller than 2.5 micrometres.

2. Ground-level ozone is a secondary pollutant.

3. A higher BOD indicates more organic pollution in water.

4. Acid rain is caused mainly by carbon monoxide.

Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 2 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Acid rain comes mainly from sulphur dioxide and nitrogen oxides.

PRACTICE

Match List-I (disease) with List-II (cause):

List-I (disease)	List-II (cause)
A. Minamata disease	1. Excess fluoride in drinking water
B. Itai-itai disease	2. Nitrates in drinking water
C. Blue baby syndrome	3. Cadmium
D. Fluorosis	4. Mercury

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) Minamata: mercury; itai-itai: cadmium; blue baby: nitrates; fluorosis: fluoride.

PRACTICE

Which one of the following statements is NOT correct?

A. The Bhopal gas tragedy was caused by methyl isocyanate.

B. India's National Air Quality Index has four categories.

C. A silence zone extends 100 metres around hospitals, schools and courts.

D. Thermal pollution reduces dissolved oxygen in water.

Answer: B. The AQI has six categories, from Good to Severe.

PRACTICE

Assertion (A): Fish thrive best in water bodies with a very high BOD.
Reason (R): Microbes use up dissolved oxygen while breaking down organic waste.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: D. A is false: high BOD means oxygen is used up, so fish suffer or die. R is true.

PRACTICE

Fishermen notice fish dying in a stretch of river just downstream from where a power plant releases its cooling water, though no chemicals are found. Which type of pollution is the most likely cause?

A. Noise pollution

B. Thermal pollution

C. Radioactive pollution

D. Light pollution

Answer: B. Hot water holds less dissolved oxygen and stresses fish: thermal pollution.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write air pollutants, sources, smog, acid rain, AQI, NCAP and GRAP. 2. Write water-quality terms, the metal diseases and soil pollution. 3. Write noise limits, thermal, radioactive, light and marine pollution, the laws and bodies, and AP's cases.

NOTEBOOK PAGE · Day 30: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Ocean water

Area	Write
Oceans	Four oceans; floor relief (6 parts); temperature, thermocline; salinity (35‰, highest, lowest, Bay vs Arabian Sea)

Movements	Waves and tsunami; tides (causes, spring, neap, Fundy, Indian tidal ports); currents (causes, warm and cold list, effects, Indian Ocean, El Niño, IOD); coral reefs

ECONOMY: Planning and NITI Aayog

- Pre-1950 plans; Planning Commission and NDC; the 12 Plans with focus and models; plan holidays and rolling plan; strategy shift after 1991; NITI Aayog structure, role, indices, programmes; Planning Commission vs NITI table; AP's aspirational districts.

SCIENCE: Pollution

Write: air pollutants (primary, secondary), sources, smog types, acid rain, AQI and NCAP, GRAP; water quality terms (BOD, DO, COD) and metal diseases; soil, noise limits, thermal, radioactive, light, marine; laws and bodies; AP cases.